

# HIKMAH CloudInfra

Architecture & Operations — how this autonomous newsletter is built and run

Domain: Infrastructure Intelligence for Platform Engineers

Auto: Friday 06:00 GST

Vol. I · Global Edition

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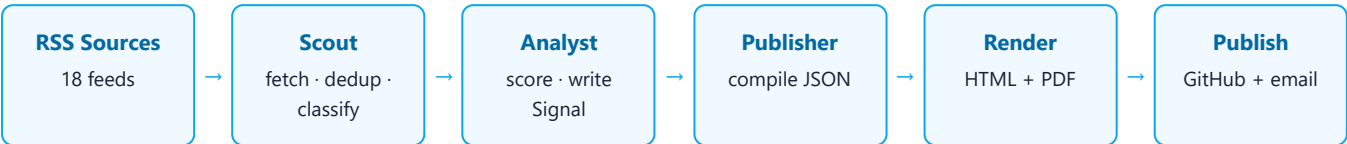
## 01 What this does

HIKMAH CloudInfra is a fully autonomous weekly intelligence newsletter. Every week a CrewAI agent crew reads 18 curated industry sources, removes anything already covered, scores what remains for a **senior audience** (engineers, architects, consultants, executives), and publishes the strongest **24 stories** — 6 in each of 4 sections — as desktop and mobile web pages and PDFs, archived on GitHub Pages.

It is not written for juniors: each story carries a decision-grade **"Signal"** that fuses the architectural implication with the strategic/business consequence — adoption timing, cost, risk, and the move a senior leader should make.

## 02 How CrewAI works here

CrewAI orchestrates a small team of role-specialised LLM agents that hand work to each other in a fixed sequence (a "crew"). Each agent has a role, a goal, a model, and optionally tools it can call. Output of one task becomes the context of the next.



## 03 The agents

| #  | AGENT                          | MODEL             | TOOLS            | RESPONSIBILITY                                                                                                                                                                        |
|----|--------------------------------|-------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | Data Scout                     | claude-haiku-4-5  | fetch_rss, dedup | Fetches every RSS feed, removes already-seen articles via SHA-256 dedup (persisted in a per-project SQLite DB), and classifies the rest into the 4 sections.                          |
| 02 | Principal Analyst & Strategist | claude-sonnet-4-6 | (reasoning only) | Scores each article 0-100 for senior-audience significance, drops anything below 65, and writes the 3-sentence summary plus the decision-grade 'Signal'. Keeps the top 6 per section. |
| 03 | Intelligence Publisher         | claude-sonnet-4-6 | (reasoning only) | Compiles the final, schema-valid JSON payload (issue metadata, stats, ticker, 4 sections x 6 entries) that drives the templates. Emits JSON only.                                     |

Models are addressed as anthropic/claude-\* so CrewAI routes them to Anthropic. The Scout uses the fast Haiku model for high-volume triage; the Analyst and Publisher use Sonnet for judgement and strict formatting.

## 04 Context & data flow

- **Tools as context:** the Scout's `fetch_rss` tool returns raw articles; its `dedup` tool checks each URL's SHA-256 against a per-project SQLite DB so a story is never repeated across issues.
- **Task chaining:** each agent's output is passed as the *context* of the next task — Scout→Analyst→Publisher — so the Analyst sees the classified set and the Publisher sees the scored set.
- **Structured payload:** the Publisher emits one JSON object — issue metadata, stats, ticker headlines, and 4 sections x 6 entries (title, url, source, date, summary, signal, keywords, vendors, score).
- **Branding merged in code:** the project's colours, wordmark and section taxonomy (from `brand.py`) are merged into the payload before a shared Jinja2 template renders it.

## 05 Coverage — the 4 sections

### 01 · Cloud Platforms

AWS · Azure · GCP · Multi-Cloud · IaC · FinOps · Serverless

**Watching:** AWS · Microsoft Azure · Google Cloud · HashiCorp · Pulumi

### 02 · Containers & Kubernetes

Docker · Kubernetes · Helm · Service Mesh · WASM · OCI

**Watching:** Docker · CNCF · Istio · Cilium · Argo · Crossplane

### 03 · Databases & Storage

PostgreSQL · MongoDB · Redis · ClickHouse · S3 · Object Storage

**Watching:** PostgreSQL · MongoDB · Redis · ClickHouse · Apache Arrow

### 04 · Edge, CDN & Telecom Cloud

Edge Compute · CDN · WASM at Edge · MEC · Telecom Cloud · vRAN

**Watching:** Cloudflare · Fastly · AWS Outposts · Azure Edge · MEC

## 06 The suite it belongs to

This is one of **five independent projects** in the HIKMAH suite, each with its own crew, sources, dedup DB, brand and weekly slot, all sharing one library (`hikmah-shared`: `renderer`, `PDF generator`, `GitHub publisher`, `email sender`, `dedup`):

| PROJECT      | DOMAIN                           | DAY (06:00 GST) |
|--------------|----------------------------------|-----------------|
| signal       | Telecom / 5G / RAN               | Tuesday         |
| intelligence | AI / Agentic / LLM               | Wednesday       |
| dataml       | MLOps / Data Science             | Thursday        |
| cloudinfra   | Cloud / Containers / Edge        | Friday          |
| dataarch     | Databases / Big Data / GPU / API | Saturday        |

## 07 What an "issue" is

Each run produces one numbered issue with **4 deliverables** — published to `CrewAI/cloudinfra/issues/` and served on GitHub Pages:

- `issue-NNN.html` — desktop edition (multi-column)
- `issue-NNN-mobile.html` — phone edition (single column)

- [issue-NNN.pdf](#) — A4 print/archive PDF
- [issue-NNN-mobile.pdf](#) — phone-format PDF

An issue number auto-increments each full run. Runs can be **manual** (`run_now.py cloudinfra`) or **automatic** (the weekly scheduler), with an offline `--demo` mode for previewing without an API key.

## 08 Sources (18)

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|--------------------------------------------|---------------------------------------|
| • <a href="#">aws.amazon.com</a>           | • <a href="#">netflixtechblog.com</a> |
| • <a href="#">azure.microsoft.com</a>      | • <a href="#">redhat.com</a>          |
| • <a href="#">cloudblog.withgoogle.com</a> | • <a href="#">thenewstack.io</a>      |
| • <a href="#">kubernetes.io</a>            | • <a href="#">grafana.com</a>         |
| • <a href="#">cncf.io</a>                  | • <a href="#">istio.io</a>            |
| • <a href="#">docker.com</a>               | • <a href="#">aws.amazon.com</a>      |
| • <a href="#">blog.cloudflare.com</a>      | • <a href="#">gremlin.com</a>         |
| • <a href="#">hashicorp.com</a>            | • <a href="#">datadoghq.com</a>       |
| • <a href="#">allthingsdistributed.com</a> | • <a href="#">elastic.co</a>          |